

Question		Marks
1(a)	N1L: An object at rest will remain <u>at rest</u> and an object in motion will remain in motion at <u>constant velocity</u> in the <u>absence of an external resultant force</u> .	B1
	N2L: The <u>rate</u> of change of momentum of a body is <u>directly</u> proportional to the <u>resultant</u> force acting on the body and occurs in the direction of the resultant force.	B1
	N3L: If body A exerts a force on body B, then body B exerts a force of the <u>same type</u> that is equal in magnitude and opposite in direction on body A.	B1
1(b)(i)	Initially, suitcase moving with same velocity as bus before braking	B1
	By N1L, suitcase will continue to move forward in the absence of external force acting on it because not secured (or small frictional force between when the suitcase was still in contact with the bus that is not sufficient to stop it from moving forward) when brakes applied	B1
	Claim invalid	A0
1(b)(ii)	The two equal and opposite forces act on <u>cart and labourer separately</u>	B1
	Cart pulled forward by the force applied by the labourer on the cart. Labourer moved forward by the <u>frictional force between feet/shoes and ground</u> .	B1
	Reasoning false	A0
1(b)(iii)	Throw bag of gold in direction away from shore	B1
	By N3L, man experiences a force equal in magnitude and opposite in direction from the force he exerted on the bag of gold. This forces accelerates the man towards shore during pushing, thereafter he will glide over ice at constant speed	B1
1(c)(i)	By N3L, force experienced by car <u>same</u> as force experienced by truck	B1
1(c)(ii)	Total mass of truck and its driver <u>larger than</u> total mass of car and its driver, hence the truck and its driver experiences smaller change in velocity over same duration of time	B1
	Safety belt exert smaller force on truck driver in stopping his forward velocity	B1
	Truck driver likely less injured than car driver	A0

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Question	Marks
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2(a)	Area under F vs graph is w.d. in stretching bow. More potential energy is	D1
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